

Introduction to Embedded Systems & IoT Device Design

Embedded Systems Task – 1

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Objective

The objective of this project is to understand the fundamentals of embedded systems and Internet of Things (IoT) by designing a simple Smart Temperature Monitoring System. This project helps in learning about microcontrollers, sensors, embedded applications, and the implementation of a basic IoT prototype.

1. Introduction to Embedded Systems

What is an Embedded System?

An embedded system is a specialized computer system designed to perform a specific task or set of tasks within a larger system. Unlike general-purpose computers, embedded systems are dedicated to a particular function and are optimized for efficiency, reliability, and real-time performance.

An embedded system typically consists of hardware components such as a microcontroller, sensors, actuators, and software programmed to perform a specific operation.

Characteristics of Embedded Systems

- Dedicated functionality
 - Real-time operation
 - Low power consumption
 - High reliability
 - Compact size
 - Cost-effective design
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Real-World Applications of Embedded Systems

Smart Home Systems

Embedded systems are used in smart lighting, smart thermostats, security cameras, and home automation devices.

Automotive Systems

Modern vehicles use embedded systems for engine control, anti-lock braking systems (ABS), airbags, cruise control, and navigation systems.

Healthcare Devices

Medical devices such as heart rate monitors, blood glucose monitors, ventilators, and pacemakers rely on embedded systems.

Robotics

Robots use embedded systems for sensing, control, movement, and communication.

Industrial Automation

Embedded systems control manufacturing equipment, monitoring systems, and industrial robots.

Consumer Electronics

Devices such as washing machines, microwave ovens, televisions, and smartphones contain embedded systems.

Difference Between Microcontroller and Microprocessor

Feature	Microcontroller	Microprocessor
Definition	Integrated chip containing CPU, memory, and I/O ports	CPU only
Memory	Built-in RAM and ROM	External memory required
Cost	Low	High
Power Consumption	Low	High
Applications	Embedded systems	Personal computers and laptops
Speed	Moderate	High

Examples

Microcontrollers:

- Arduino UNO (ATmega328P)

- ESP32
- Raspberry Pi Pico

Microprocessors:

- Intel Core i5
 - Intel Core i7
 - AMD Ryzen Processors
-

2. Component Study

Microcontrollers

Arduino UNO

Arduino UNO is one of the most popular microcontroller development boards based on the ATmega328P microcontroller.

Features

- Operating Voltage: 5V
- Digital I/O Pins: 14
- Analog Inputs: 6
- Flash Memory: 32 KB
- Clock Speed: 16 MHz

Applications

- Robotics
 - Home automation
 - Sensor interfacing
 - Educational projects
-

ESP32

ESP32 is a powerful microcontroller with built-in Wi-Fi and Bluetooth capabilities.

Features

- Dual-core processor
- Wi-Fi connectivity

- Bluetooth support
- Low power consumption

Applications

- IoT devices
 - Smart home systems
 - Wireless monitoring systems
-

Raspberry Pi Pico

Raspberry Pi Pico is a low-cost microcontroller board based on the RP2040 chip.

Features

- Dual-core ARM Cortex-M0+
- 264 KB SRAM
- Programmable I/O
- Low power operation

Applications

- Embedded systems
 - Automation projects
 - Sensor-based applications
-

Sensors

Temperature Sensor

A temperature sensor measures the surrounding temperature and converts it into an electrical signal.

Example

TMP36 Temperature Sensor

Applications

- Weather monitoring
- Industrial monitoring
- Smart homes

Motion Sensor

A motion sensor detects movement within its sensing range.

Example

PIR (Passive Infrared) Sensor

Applications

- Security systems
- Automatic doors
- Smart lighting

Light Sensor

A light sensor measures the intensity of light.

Example

LDR (Light Dependent Resistor)

Applications

- Automatic street lights
- Smart lighting systems
- Solar tracking systems

3. Mini Project: Smart Temperature Monitoring System

Project Description

The Smart Temperature Monitoring System continuously measures environmental temperature using a temperature sensor. When the temperature exceeds a predefined threshold value, an LED and buzzer are activated to alert the user.

This system can be implemented using real hardware or simulated using Tinkercad Circuits.

Components Required

- Arduino UNO
 - TMP36 Temperature Sensor
 - LED
 - 220Ω Resistor
 - Piezo Buzzer
 - Breadboard
 - Jumper Wires
-

Circuit Connections

TMP36 Temperature Sensor

- Left Pin → 5V
- Middle Pin → A0
- Right Pin → GND

LED

- Anode (+) → Arduino Pin 13 through 220Ω resistor
- Cathode (-) → GND

Buzzer

- Positive Terminal → Arduino Pin 8
 - Negative Terminal → GND
-

Arduino Program

```
// Smart Temperature Monitoring System
```

```
// Arduino UNO + TMP36 Sensor + LED + Buzzer
```

```
int tempPin = A0;
```

```
int ledPin = 13;
```

```
int buzzerPin = 8;
```

```
float threshold = 30.0;
```

```
void setup()
```

```
{
```

```
  Serial.begin(9600);
```

```
  pinMode(ledPin, OUTPUT);
```

```
  pinMode(buzzerPin, OUTPUT);
```

```
}
```

```
void loop()
```

```
{
```

```
  int sensorValue = analogRead(tempPin);
```

```
  float voltage = sensorValue * (5.0 / 1023.0);
```

```
  float temperature = (voltage - 0.5) * 100;
```

```
  Serial.print("Temperature: ");
```

```
  Serial.print(temperature);
```

```
  Serial.println(" C");
```

```
  if (temperature > threshold)
```

```
  {
```

```
    digitalWrite(ledPin, HIGH);
```

```
    tone(buzzerPin, 1000);
```

```
  }
```

```
  else
```

```
  {
```

```
    digitalWrite(ledPin, LOW);
```

```
    noTone(buzzerPin);
```

```
}
```

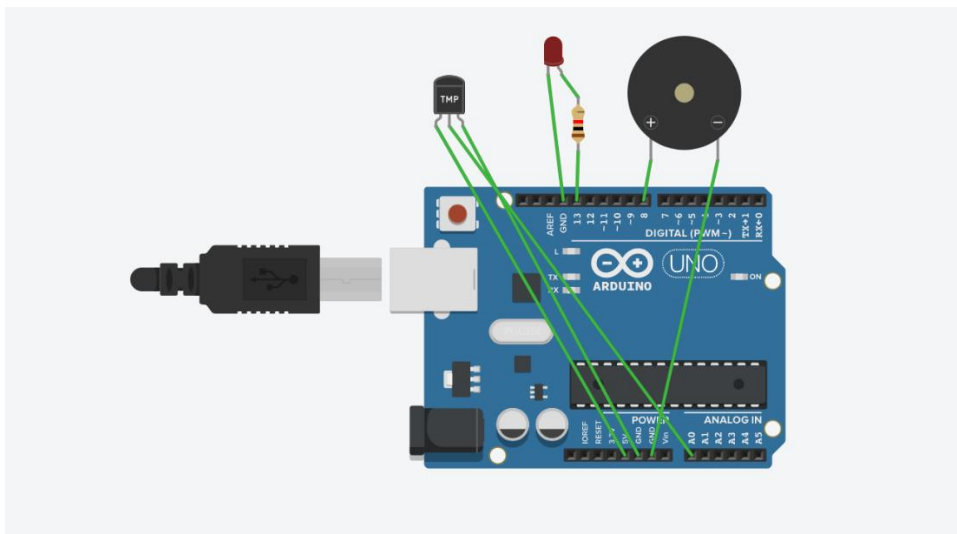
```
delay(1000);
```

```
}
```

4. Working Principle

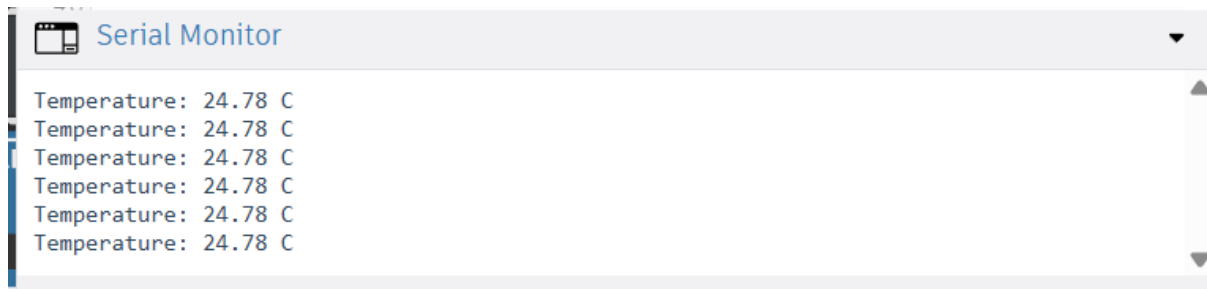
1. The TMP36 sensor continuously measures the surrounding temperature.
 2. Arduino reads the analog signal from the sensor.
 3. The analog value is converted into temperature in degrees Celsius.
 4. The measured temperature is displayed on the Serial Monitor.
 5. If the temperature exceeds the threshold value of 30°C:
 - LED turns ON.
 - Buzzer produces an alarm sound.
 6. If the temperature is below the threshold:
 - LED remains OFF.
 - Buzzer remains OFF.
 7. The process repeats every second for continuous monitoring.
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5. Circuit Diagram



Software used: Tinkercad

Output



Link for the mini project

https://www.tinkercad.com/things/aaxJ1hSFMjz-smart-temperature-monitoring-system?sharecode=zUsk_U60fC8w-Jk-QU0iYZEukppBklmHXvkD9AB1Hxk

6. Real-World Applications

Smart Homes

Monitoring room temperature and generating alerts during extreme temperatures.

Industrial Safety

Monitoring equipment temperature to prevent overheating.

Server Rooms

Detecting high temperatures that could damage servers.

Greenhouses

Maintaining optimal temperature conditions for plant growth.

Healthcare Equipment

Monitoring temperature-sensitive medical devices.

7. Advantages

- Low-cost implementation
 - Easy to design and maintain
 - Real-time monitoring
 - High reliability
 - Expandable for IoT applications
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Conclusion

Embedded systems play a vital role in modern technology by providing intelligent and automated solutions. In this project, a Smart Temperature Monitoring System was designed using Arduino UNO and a temperature sensor. The system continuously monitors temperature and provides alerts whenever the temperature exceeds a specified threshold. This project demonstrates the practical application of embedded systems and IoT concepts and serves as a foundation for developing more advanced smart monitoring systems.
